

Unit P1: Pure Mathematics 1

Compulsory unit for IAS Mathematics and Pure Mathematics Compulsory unit for IAL Mathematics and Pure Mathematics

Externally assessed

P1.1 Unit description

Algebra and functions; coordinate geometry in the (x,y) ; trigonometry; differentiation; integration.

P1.2 Assessment information

1. Examination

- First assessment: January 2019.
- The assessment is 1 hour and 30 minutes.
- The assessment is out of 75 marks.
- Students must answer all questions.
- Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*.
- The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.

2. Notation and formulae

Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.

Formulae that students are expected to know are given below and will **not** appear in the booklet; *Mathematical Formulae and Statistical Tables*. This booklet will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Quadratic equations

$$ax^2 + bx + c = 0 \text{ has roots } \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Trigonometry

In the triangle ABC ,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\text{area} = \frac{1}{2} ab \sin C$$

$$\text{arc length} = r\theta$$

$$\text{area of sector} = \frac{1}{2} r^2 \theta$$

P1.3 Unit content

What students need to learn:	Guidance
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